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Spacecraft Attitude Dynamics

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Project specifications:
This table is mandatory

	Assigned specification	**Modifications (if any)**	**Motivation for modification**
Platform	Cubesat 6U		
Attitude parameters	Direction Cosines		
Mandatory sensor	Magnetic field sensor		
Actuators	3 Reaction Wheels		

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1 Introduction

The following report analyses the design and simulation of an Attitude Determination and Control System (ADCS) for a 6U CubeSat undertaking a Sun-pointing mission in Low Earth Orbit (LEO). The spacecraft moves along a circular, highly inclined retrograde orbit. The orbit is assumed to maintain constant orbital parameters over the entire simulation time.

The mission is divided into three stages: detumbling, slew manoeuvre, and pointing. The control torque is generated using three reaction wheels and a tri-axial magnetic torquer. During detumbling, the torque command is computed through an angular-rate damping method, whereas in slew manoeuvre and pointing phases, a proportional-derivative (PD) approach is used.

The attitude parameter used for simulating the kinematics of the satellite is the direction cosine matrix (DCM) expressing the relative orientation between the body frame of the spacecraft and the Earth-centred inertial (ECI) reference frame. Attitude determination also uses the DCM and is achieved through an algebraic algorithm, exploiting the information coming from a magnetometer and a Sun sensor.

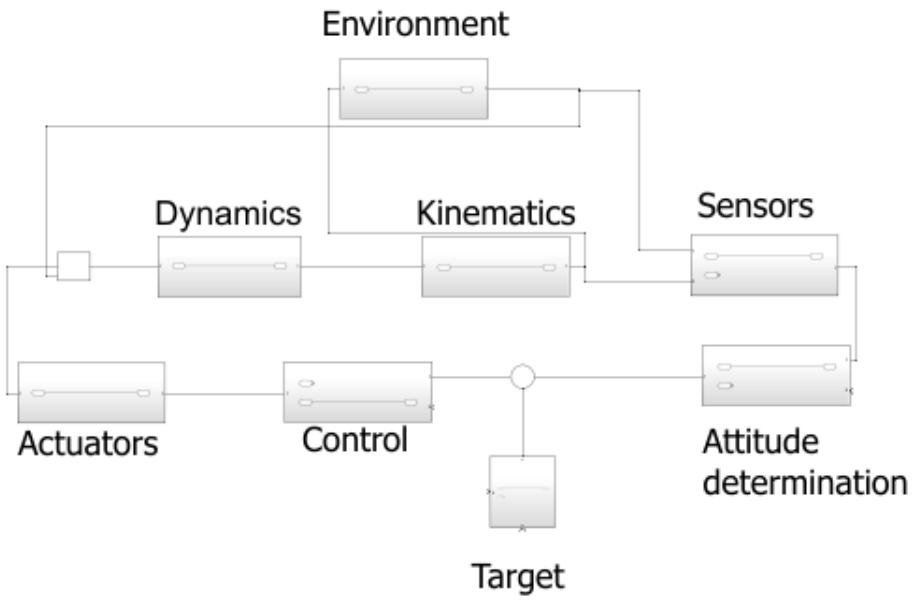


Figure 1: Block scheme of the complete model

2 Mission Description

The orbital parameters, taken from a real mission [1], are the following:

- Semimajor axis $a = 6978\,\text{km}$
- Eccentricity $e = 0$

- Inclination $i = 97.8\,\text{deg}$
- Right ascension of the ascending node $\Omega = 0\,\text{deg}$
- Argument of the pericenter $\omega = 0\,\text{deg}$
- Initial true anomaly $\theta_0 = 0\,\text{deg}$

Since the objective of our mission is pointing the Sun, this orbit has been chosen so that the spacecraft never enters eclipse.

Our target reference frame is built in the following way. The x axis corresponds to the Satellite-Sun direction. The y axis is given by the cross product between the x axis and the $[0; 0; 1]$ vector. The z axis is derived consequently, in order to obtain a right-handed orthonormal frame.

The aim of our mission is to maintain a pointing error below 5 degrees during the pointing phase. To be more specific, we want to keep the angle between the x axis of the target reference frame and the x axis of the body frame below 5 degrees.

2.1 Mission Phases

As already stated, our mission is divided into three phases: detumbling, slew manoeuvre and pointing (also called tracking).

At the beginning of the simulation, the spacecraft is characterized by a random but large angular velocity, with a magnitude typically ranging from 5 degrees per second to 40 degrees per second. During the detumbling phase, we aim to drastically diminish the angular spin rate of the spacecraft.

Once the detumbling phase is done and the target angular velocities are reached, the slew manoeuvre phase will take place. Here, our objective is to bring the body frame to coincide almost exactly with the target reference frame.

The final tracking phase aims to make the satellite maintain the reference attitude, so that pointing requirements are satisfied.

2.2 Satellite's properties

The CubeSat is modelled as a rectangular prism with dimensions $30 \times 20 \times 10\,\text{cm}$, with a mass of $7.23\,\text{kg}$. It is characterized by the following inertia matrix:

$$\mathbf{I} = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}$$

$$I_{xx} = 0.0301\,\text{kg}\,\text{m}^2, \quad I_{yy} = 0.0603\,\text{kg}\,\text{m}^2, \quad I_{zz} = 0.0783\,\text{kg}\,\text{m}^2.$$

Principal inertia moments were computed assuming a uniform mass distribution over the volume and taking a reference frame that is aligned with the edges of the parallelepiped, with the origin in the center of gravity.

3 Dynamics & Kinematics Modelling

The dynamics of the system are derived from the Euler equations (1), which describe the rotational dynamics of a rigid body.

$$\mathbf{I}\dot{\underline{\omega}} + \underline{\omega} \wedge \mathbf{I}\underline{\omega} = \underline{\tau} \tag{1}$$

$\underline{\omega}$ is the angular velocity vector expressed in body frame, $\underline{\tau}$ is the vector of torques acting on the system. The kinematics are instead modelled using a direction cosine matrix (DCM) expressing the relative orientation between the body frame of the spacecraft and the Earth-centred inertial (ECI) reference frame. The evolution of the DCM is computed by integrating the differential set of equations given by the following system:

$$\frac{d\mathbf{A}_{B/N}}{dt} = -\left[\underline{\omega}\wedge\right]\mathbf{A}_{B/N}(t) \tag{2}$$

where $[\underline{\omega}\wedge]$ represents the skew-symmetric matrix associated with the angular velocity vector.

To reduce the impact of numerical errors, at each timestep we apply the following equation to keep $\mathbf{A}_{B/N}$ orthonormal:

$$\mathbf{A}_{B/N}(t) = \frac{3}{2}\mathbf{A}_0(t) - \frac{1}{2}\mathbf{A}_0(t)\,\mathbf{A}_0^{\mathrm{T}}(t)\,\mathbf{A}_0(t) \tag{3}$$

To validate this part of the model, the conservation of angular momentum and kinetic energy was checked, under the assumption of no external torques applied to the system. Moreover, we checked that the attitude of the satellite evolved in time consistently with known cases, as shown in Fig.2, where the equilibrium configuration $\underline{\omega} = [0; 0; \omega_z]$ was chosen as initial condition.

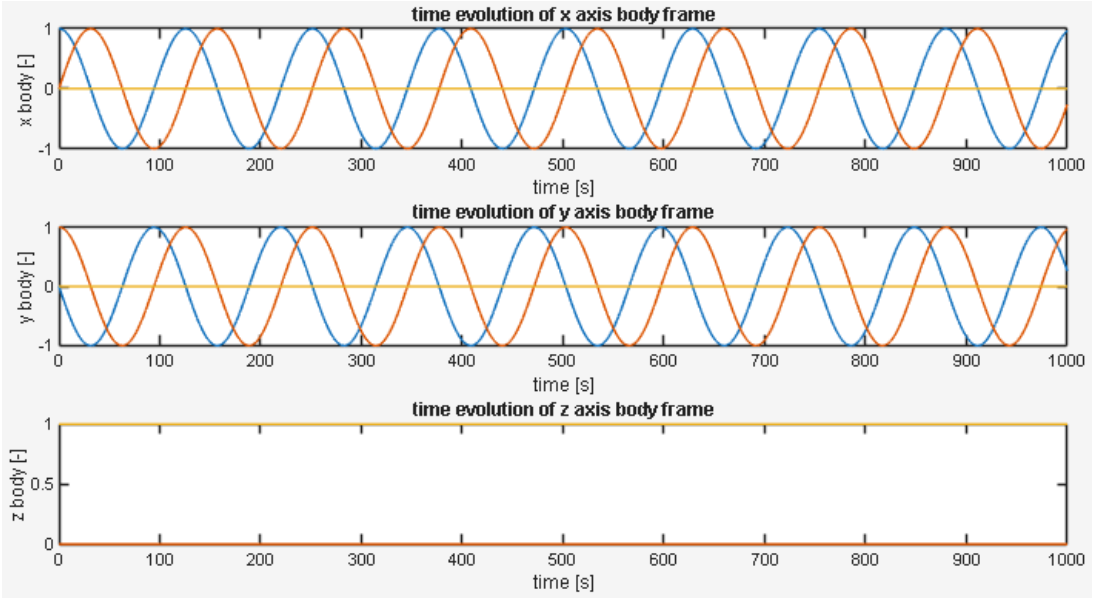


Figure 2: Evolution in time of the orientation of the three body axes of the satellite, with an initial angular velocity of $[0; 0; \omega_z]$

4 Environment Modelling

4.1 Disturbance torques

Using the relationships expressed in [2], we estimated the maximum possible torque generated by each disturbance:

- Gravity gradient $\tau_{max} = \frac{3GM_t}{2r^3}|I_{max} - I_{min}| = 8.4817 \cdot 10^{-8}\,\mathrm{N\,m}$
- Air drag $\tau_{max} = \frac{1}{2}\,\rho\,V^2\,A_s\,C_D\,(c_{pa} - c_g) = 7.7550 \cdot 10^{-8}\,\mathrm{N\,m}$
- Solar radiation pressure $\tau_{max} = P_s A_s (1+q)\,(c_{ps} - c_g) = 6.6512 \cdot 10^{-8}\,\mathrm{N\,m}$
- Magnetic field $\tau_{max} = D_s\,B_{\max} = 8.6603 \cdot 10^{-7}\,\mathrm{N\,m}$

where

- GM_t is the gravitational parameter of the earth, which is equal to $398600\,\mathrm{km^3 s^{-2}}$.
- $r = 6978\,\mathrm{km}$ is the distance going from the spacecraft to the centre of the Earth.
- V is the maximum magnitude of the spacecraft's velocity. Since the orbit is circular, V has been calculated using the relationship $v = \sqrt{\frac{GM_t}{r}}$. Consequently, V has been posed equal to $7557\,\mathrm{m\,s^{-1}}$.
- $\rho = 1 \cdot 10^{-13}\,\mathrm{kg\,m^{-3}}$ is the air density at 600 km of altitude.
- C_d has been assumed equal to 2.2.
- $A_s = 6 * 10^{-2}\,\mathrm{m^2}$ is the largest external surface surface of the CubeSat.
- P is the solar radiation pressure, which is equal to $4.56 * 10^6\,\mathrm{Nm^{-2}}$
- q has been assumed equal to 0.3.
- Both distances $c_{ps} - c_g$ and $c_{pa} - c_g$ have been calculated assuming that c_g is at the geometrical center of the rectangular prism. With c_{ps} and c_{pa} being in one of the edges of the spacecraft. Consequently, both distances have been imposed equal to $18.7 * 10^{-2}\,\mathrm{m}$.
- By propagating the orbit, we have been able to verify that $B_{max} = 5 * 10^{-5}\,\mathrm{T}$.
- Based on the information provided in [3], we made the following assumption: $D_{max} = 0.0173\,\mathrm{Am^2}$.

Therefore, we decided to neglect the effects of the air drag and of the solar radiation pressure, modelling only the gravity gradient and the geomagnetic field disturbances.

4.1.1 Gravity Gradient

The gravity gradient torque is computed as follows:

$$\begin{cases} \tau_x = \dfrac{3GM_t}{r^3}\left(I_z - I_y\right)c_2 c_3 \\ \tau_y = \dfrac{3GM_t}{r^3}\left(I_x - I_z\right)c_1 c_3 \\ \tau_z = \dfrac{3GM_t}{r^3}\left(I_y - I_x\right)c_2 c_1 \end{cases} \tag{4}$$

where $\underline{c}$ is defined as:

$$\underline{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \mathbf{A}_{B/L} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \tag{5}$$

where $\mathbf{A}_{B/L}$ is the matrix representing the rotation from the local vertical local horizontal (LVLH) frame to the body frame and it is derived from the knowledge of $\mathbf{A}_{B/N}$ (representing the rotation from the inertial frame to the body frame) and $\mathbf{A}_{L/N}$ (representing the rotation from inertial frame to LVLH frame) which depends exclusively on the true anomaly of the Keplerian orbit, that in turn depends only on time.

The gravity gradient model has been validated by comparison with known theoretical cases. As shown in Fig.3, in a circular equatorial orbit with initial conditions $\underline{\omega} = [0; 0; n]$ (where n is the mean angular motion of the satellite around the orbit) and $\mathbf{A}_{B/N}$ equal to $\mathbf{A}_{L/N}$, the torque generated by the gravity gradient goes to zero because the body axes of the satellite are always aligned with the LVLH frame.

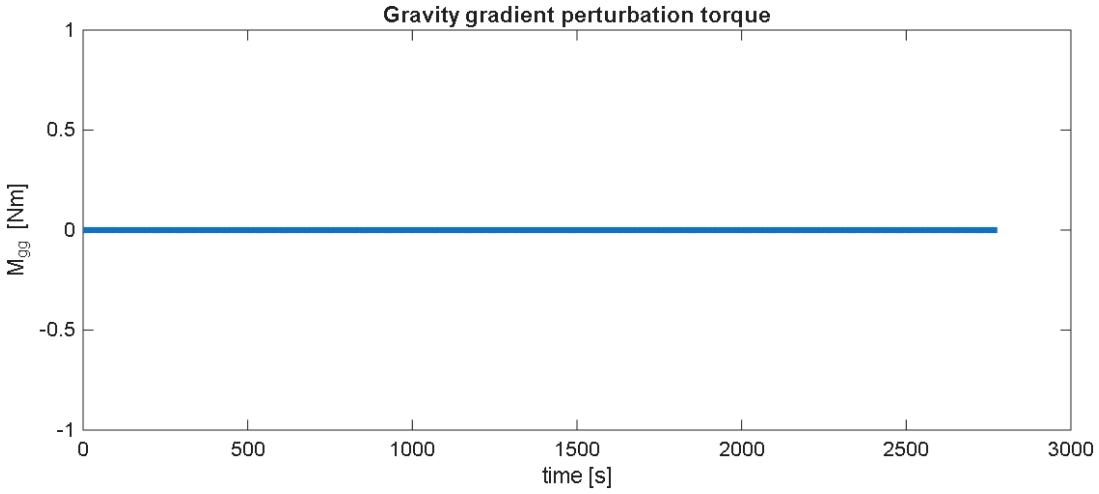


Figure 3: Torque acting on the satellite due to gravity gradient perturbation in a circular equatorial orbit, with initial angular velocity $\underline{\omega} = [0; 0; n]$

4.1.2 Magnetic torque

As shown in [3], it is reasonable to model the residual magnetic dipole moment of a 6U CubeSat within the range of 10^{-3}–$10^{-1}\,\mathrm{A\,m^2}$. Accordingly, in this project the residual dipole moment vector $\underline{m}_{res}$ is assumed to be equal to $[0.01, 0.01, 0.01]\,\mathrm{A\,m^2}$ in body frame.

The geomagnetic field has been modelled using a spherical harmonic expansion up to degree 11. From the integration of motion of the CubeSat, and under the hypothesis of an initial right ascension of Greenwich equal to zero, we are able to determine both the latitude and the longitude of the satellite. Therefore, we can compute the components of the magnetic field with respect to the ECI frame. And using $A_{B/N}$, we derive the magnetic field $\underline{B}$ expressed in body frame. Consequently, the disturbance torque is computed as

$$\underline{\tau}_{dist} = \underline{m}_{res} \wedge \underline{B} \tag{6}$$

The geomagnetic field model was validated by confirming that the computed magnetic field magnitude is consistent with the contour levels reported in [4]. The table below compares the magnitude of the magnetic field obtained through our model with the one in [4], for different coordinates and order of the model.

order	**altitude**	**latitude**	**longitude**	B_{model}	$B_{theorical}$
13	500 km	$0°$	$0°$	24259 nT	$\approx$ 24000 nT
4	500 km	$60°S$	$120°W$	40111 nT	$\approx$ 40000 nT
13	1500 km	$60°N$	$0°$	27972 nT	$\approx$ 28000 nT
4	1500 km	$20°S$	$120°E$	26149 nT	$\approx$ 26000 nT
13	8000 km	$0°$	$30°W$	2354 nT	$\approx$ 2400 nT
2	8000 km	$60°N$	$120°E$	4853 nT	$\approx$ 4800 nT

After further analysis, we have concluded that small differences between B_{model} and $B_{theorical}$ are primarily caused by the time evolution of the Gauss coefficients.

4.2 Orbit propagation

To determine the Sun–Earth–satellite geometry, an ideal Keplerian model (two-body dynamics) was adopted. The following motions were numerically propagated:

- the satellite's orbit around the Earth
- the Earth's orbit around the Sun

Propagation has been done by integrating the equation of motion

$$\ddot{\underline{r}} = -\mu \frac{\underline{r}}{\|\underline{r}\|^3} \tag{7}$$

from which we obtain the position vectors $\underline{r}_{SC/E}$ (satellite with respect to the Earth) and $\underline{r}_{E/S}$(Earth with respect to the Sun), both expressed in a common inertial reference frame. By combining the two, the vector

$$\underline{r}_{S/SC} = \underline{r}_{S/E} - \underline{r}_{SC/E}. \tag{8}$$

has been computed, providing the relative direction and distance between the satellite and the Sun.

5 Sensors

Sensors represent the interface between the external physical world and the digital world of the on-board computer. They sense a physical quantity and transform it into an electrical signal, so that the attitude determination system can process it to retrieve the orientation of the spacecraft. In our case, we are given a magnetometer as mandatory sensor, which provides a measurement of the magnitude and direction of the magnetic field. It really suits our scenario well, since it is very light and works well in LEO. However, it is not enough to derive the complete attitude, since it produces only a vector, while at least two are needed. For this reason, it was decided to add a Sun sensor, which provides the direction of the Sun. The choice was guided by the objective of the mission, but also by the good accuracy and low mass of this kind of sensor.

5.1 Magnetic sensor

The model of the magnetometer is based on the information reported in the datasheet [5]. Specifically, a performance model was implemented, accounting for different kinds of non idealities:

- Non orthogonality error has been modelled as a lower triangular matrix that multiplies the magnetic field vector.
- White noise has been modelled with the normal random number generator (null mean and unitary variance) multiplied by a gain equal to $\frac{D}{\sqrt{2T_s}}$ [1].
- Pink noise has been modelled as the white noise plus a discrete filter with an attenuation of 3 dB/oct.
- Brown noise has been modelled as the white noise plus a first order lowpass filter with an attenuation of 6 dB/oct.
- Quantisation error takes into account the resolution of the sensor (6.5 nT)
- Saturation takes into account the measurement range (-50000 nT to $+\,50000$ nT)

The sensor works at a frequency of 10 Hz, meaning the sampling time is $T_s = 0.1$ s. Exploiting this information, we modelled the temporal discretization with a zero-order hold, while the delay was set to one sampling time. Fig.4 shows the difference between the real magnetic field and the measured one, in a free motion condition, computed as $B_{err} = \|\underline{B}_{real} - \underline{B}_{measured}\|$.

5.2 Sun sensor

The Sun sensor model is based on the datasheet [6], which reports only the accuracy as performance parameter. Therefore, we are assuming that this parameter includes all the other sources of errors (noises, bias, etc.). Accuracy has been modelled through a rotation matrix that rotates the Sun direction by a small angle. As done before, delay and temporal discretization have been modelled too, while saturation has been removed since we are assuming that our spacecraft has a field of view of 360 degrees. Fig.5 shows the error affecting the measurement of the Sun direction, calculated as $err = \arccos(\frac{\underline{S}_{real} \cdot \underline{S}_{measured}}{\|\underline{S}_{real} \cdot \underline{S}_{measured}\|})$.

[1] D is the noise density and T_s is the sampling time, both taken from datasheet [5].

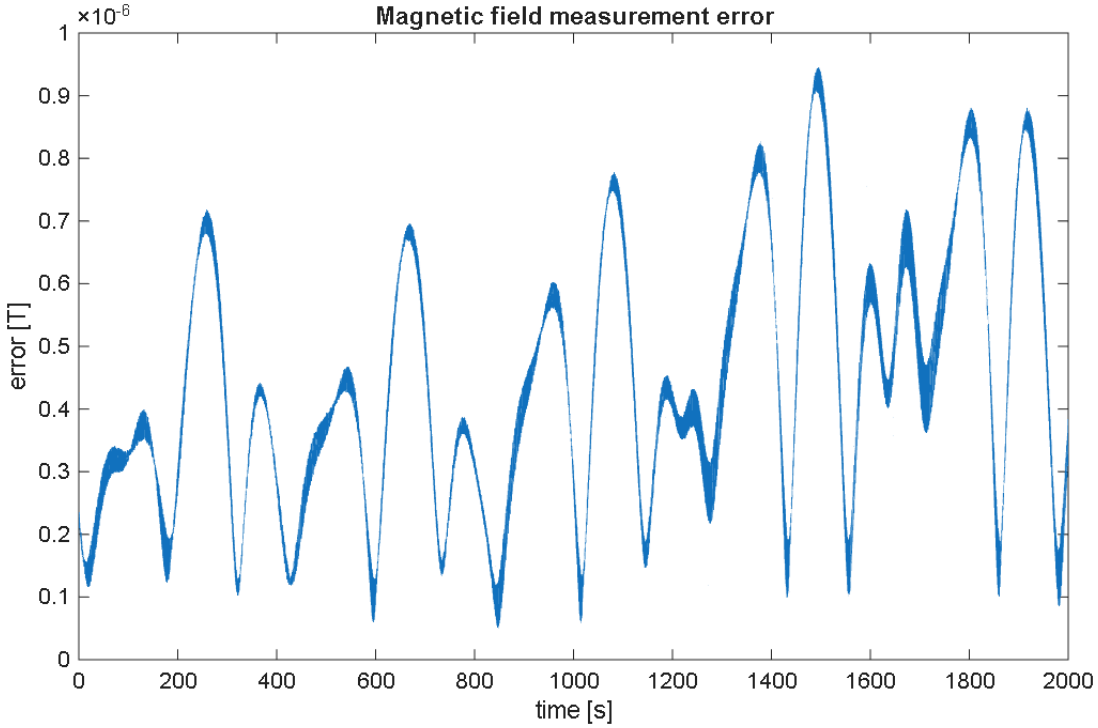


Figure 4: Error between the measured and real magnetic field.

6 Attitude determination

The attitude determination system uses the data from the sensors to derive the attitude and angular velocity of the spacecraft. Several algorithms are available, depending on the precision and speed we want to achieve. Since in the kinematics we used the direction cosine matrix, we decided, for simplicity, to keep this attitude parameter also for this part, despite knowing that using quaternions could lead to a faster algorithm. Our sensors provide two independent measurements: the direction of the Sun and the direction and magnitude of Earth's magnetic field, both in the spacecraft body frame. They are enough to compute the attitude.

6.1 Algebraic method

We first modelled the attitude determination through the algebraic method: after normalizing the magnetic field measurement, we can exploit the cross product to obtain the axis perpendicular to the plane in which the Sun vector and the magnetic field vector lie:

$$\underline{s}_1 = \frac{\underline{B}_{measured}}{\|\underline{B}_{measured}\|} \tag{9}$$

$$\underline{s}_2 = \frac{\underline{s}_1 \wedge \underline{S}_{direction}}{\|\underline{s}_1 \wedge \underline{S}_{direction}\|}. \tag{10}$$

Another cross product allows us to find the third axis of the orthonormal frame

$$\underline{s}_3 = \underline{s}_1 \wedge \underline{s}_2. \tag{11}$$

With these three unit vectors as columns we build the matrix $\mathbf{S} = [\underline{s}_1 \ \ \underline{s}_2 \ \ \underline{s}_3]$. Repeating the procedure for the magnetic field vector and the Sun vector in inertial space, we obtain

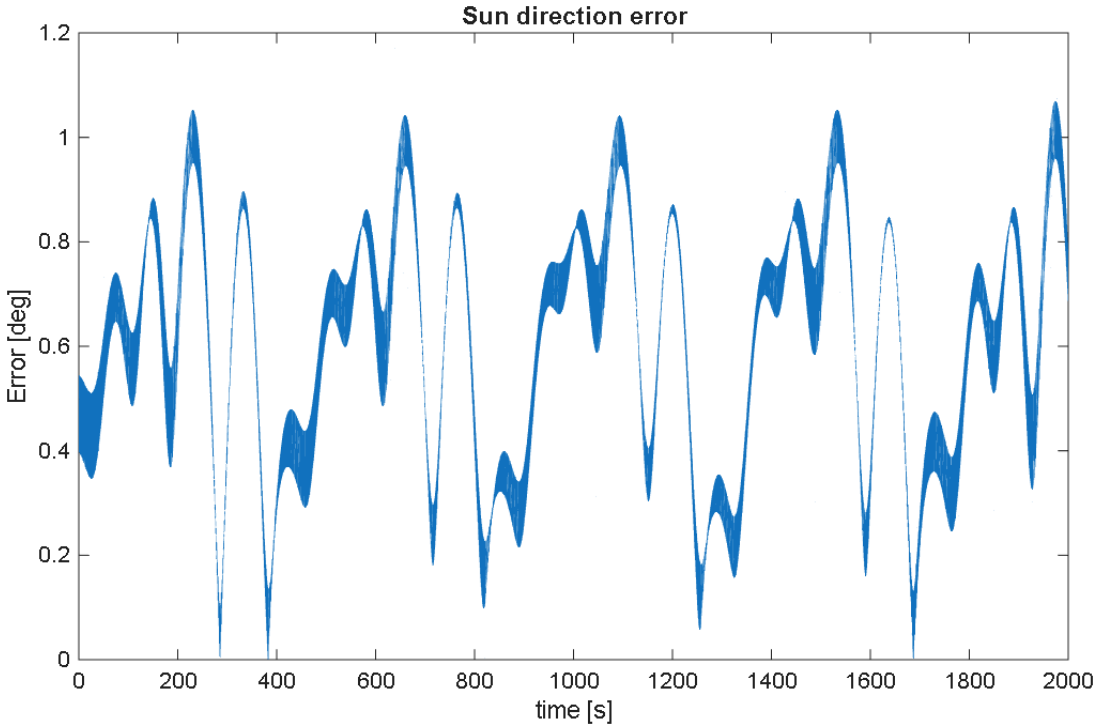


Figure 5: Error between the measured and real Sun direction

the matrix $\mathbf{V} = [\underline{v}_1 \ \underline{v}_2 \ \underline{v}_3]$. Then, just a simple matrix multiplication is needed to obtain the estimated attitude

$$\mathbf{A}_{B/N} = \mathbf{S}\mathbf{V}^{\mathrm{T}}. \tag{12}$$

The validation of this method has been done by plugging in as input the real magnetic field and Sun vectors, and then checking if the error was small.

6.2 Statistical method

Since the algebraic method does not account for the errors related to the sensors, we also modelled the statistical method, aiming to compare the results. In this case the solution is

$$\mathbf{A}_{B/N} = \mathbf{U}\mathbf{M}\mathbf{V}^{\mathrm{T}} \tag{13}$$

where the matrix M is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & det(\mathbf{U}) \cdot det(\mathbf{V}) \end{bmatrix}$$

and the matrices U and V come from the singular value decomposition of B

$$\mathbf{B} = \mathbf{U}\mathbf{\Sigma}^{\mathrm{T}}\mathbf{V}^{\mathrm{T}}. \tag{14}$$

B is computed as

$$\mathbf{B} = \sum_{i=1}^{2} \alpha_i \underline{s}_i \underline{v}_i^{\mathrm{T}} \tag{15}$$

where α_i is the relative precision of the sensor and $\underline{s}_i$ and $\underline{v}_i$ are the unit vectors already used in the algebraic method. The value of α relative to the Sun sensor has been chosen

to be equal to the accuracy value available in the datasheet. The relative precision of the magnetometer, instead, has been taken from the lecture slides, since there is no direct information in the datasheet. The value happens to be the same for both, so $\alpha_1 = \alpha_2 = 0.5$. Since the comparison in Fig.6 shows very similar results, we chose to

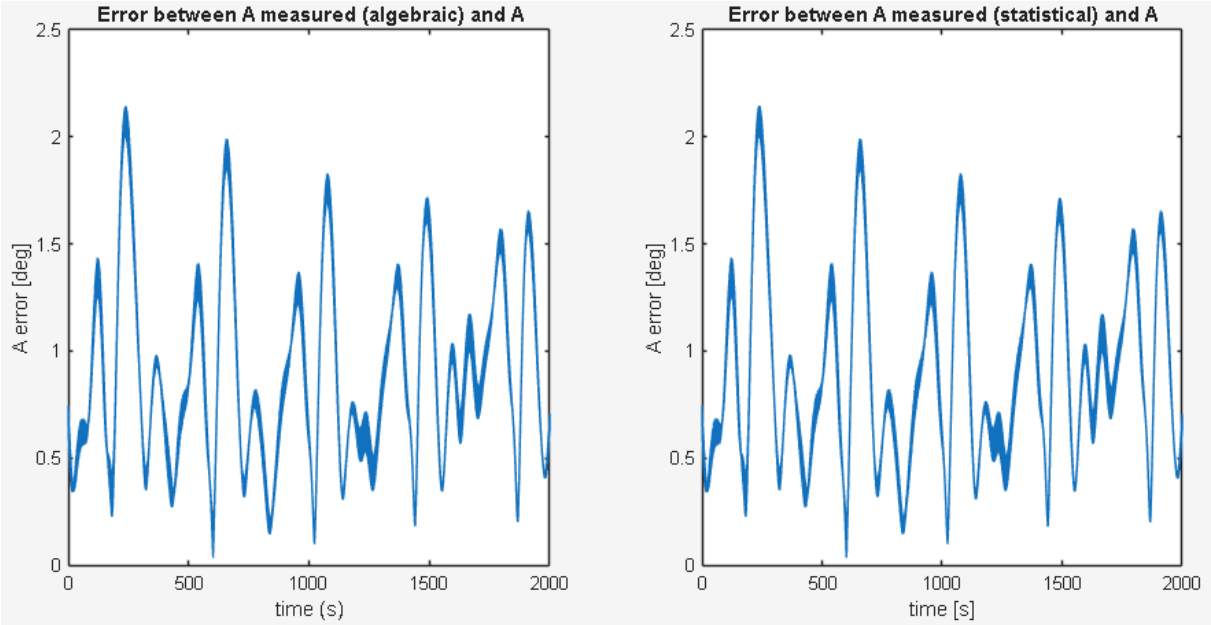


Figure 6: Comparison between algebraic and statistical method for attitude determination in a free motion situation.

adopt the algebraic method because it is less computationally demanding and therefore faster.

6.3 Angular velocity estimation

Since the satellite is not equipped with a gyroscope, on board there is no sensor capable of providing a direct measurement of the angular velocity. However, it can be estimated by inverting the kinematics

$$[\underline{\omega}\wedge] = -\dot{\mathbf{A}}_{B/N}\mathbf{A}_{B/N}{}^{\mathrm{T}}. \tag{16}$$

In the model, $\mathbf{A}_{B/N}$ is the direction cosine matrix estimated by the attitude determination, while $\dot{\mathbf{A}}_{B/N}$ has been calculated with a backward finite difference approximation. To reduce the error, the skew-symmetric components of the $[\underline{\omega}\wedge]$ matrix have been summed and divided by half. The method just described has higher accuracy the lower is the angular velocity of the satellite, since the finite difference method to estimate $\dot{\mathbf{A}}_{B/N}$ becomes less accurate the more two following configurations of the satellite are different.

7 Control Logic

As already stated, the control torque is generated using three reaction wheels and a tri-axial magnetic torquer. To avoid saturation of the reaction wheels, the control torque during detumbling is generated only by the magnetic torquer. Detumbling is considered to be over when the norm of the angular velocity of the spacecraft is brought to 5 degrees

per second. In the last two stages (slew manoeuvre and tracking), the magnetic torquer is deactivated, and the reaction wheels provide the full control torque. We consider the slew manoeuvre to be concluded when the pointing error is brought down to 5 degrees, and every component of the angular velocity vector does not exceed 2 degrees per second in absolute value. Our aim is to maintain the pointing error below 5 degrees after the slew maneouver is over.

7.1 Detumbling

During detumbling, the control torque is computed through an angular-rate damping method. Being more formal, this method is mathematically expressed as:

$$\underline{\tau} = -k\ \underline{\omega} \tag{17}$$

where $\underline{\omega}$ is the measured angular velocity vector, which is derived from the information given by the sensors. While k is a positive scalar chosen so that the detumbling is concluded within one orbit. Through a manual tuning method, we've decided to assign to k a value of 0.0007.

7.2 Slew manoeuvre and pointing

During these two phases, the control logic follows a proportional-derivative (PD) approach. We define the state vector $\underline{x}$ of the system as a six elements column vector. The first three components of it corresponds respectively to the a_{23}, a_{31} and a_{12} components of the measured DCM going from the target reference frame to the body frame. While the last three elements of $\underline{x}$ corresponds to the components of the measured angular velocity vector. The control torque is therefore calculated as:

$$\underline{\tau} = -\mathbf{K}\ \underline{x} \tag{18}$$

where the square matrix $\mathbf{K}$ is computed using the Riccati equation. To solve this equation, we defined the matrices $\mathbf{Q}$ and $\mathbf{R}$ in the following way:

$$\mathbf{Q} = \mathrm{diag}\left(\frac{1}{x_{i,\max}^2}\right) \tag{19}$$

$$\mathbf{R} = \mathrm{diag}\left(\frac{1}{u_{i,\max}^2}\right). \tag{20}$$

Since the objective of our mission is to keep the pointing error below 5 degrees, we impose the first three elements of $x_{i,max}$ to be equal to 0.087 radians (that is, 5 degrees). Analogously, the last three elements of $x_{i,max}$ are set to the maximum allowable values of the angular-rate components for the tracking phase. Therefore, these elements are imposed to be equal to 0.035 radians per second (that is, 2 degrees per second). For $u_{i,max}$ we decided, through empirical tuning, to set all its components equal to $10^{-4}\,\mathrm{N\,m}$. We have in fact noted that, in this way, pointing requirements were satisfied for all tested representative initial conditions, without saturating the reaction wheels.

7.2.1 Controllability and observability

Since our target reference frame is almost inertial, the dynamic of the system in state space representation has the following form:

$$\begin{cases} \dot{\underline{x}} = \mathbf{A}\underline{x} + \mathbf{B}\underline{u} \\ \underline{y} = \mathbf{C}\underline{x} + \mathbf{D}\underline{u} \end{cases} \tag{21}$$

Being more precise, we have:

$$\begin{cases} \begin{bmatrix} \dot{\alpha}_x \\ \dot{\alpha}_y \\ \dot{\alpha}_z \\ \ddot{\alpha}_x \\ \ddot{\alpha}_y \\ \ddot{\alpha}_z \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \alpha_x \\ \alpha_y \\ \alpha_z \\ \dot{\alpha}_x \\ \dot{\alpha}_y \\ \dot{\alpha}_z \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{1}{I_x} & 0 & 0 \\ 0 & \frac{1}{I_y} & 0 \\ 0 & 0 & \frac{1}{I_z} \end{bmatrix} \begin{bmatrix} M_x \\ M_y \\ M_z \end{bmatrix} \\ \underline{y} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \alpha_x \\ \alpha_y \\ \alpha_z \\ \dot{\alpha}_x \\ \dot{\alpha}_y \\ \dot{\alpha}_z \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} M_x \\ M_y \\ M_z \end{bmatrix} \end{cases}$$

(22)

where $[\alpha_x \ \alpha_y \ \alpha_z \ \dot{\alpha}_x \ \dot{\alpha}_y \ \dot{\alpha}_z]^{\mathrm{T}}$ is the state vector $\underline{x}$, and $[M_x \ M_y \ M_z]^{\mathrm{T}}$ is the input vector $\underline{u}$.

From this, we can derive the matrix $\mathbf{Q}$:

$$\mathbf{Q} = \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \mathbf{A^2B} & \cdots & \mathbf{A}^{n-1}\mathbf{B} \end{bmatrix} \tag{23}$$

where n represents the number of states of the system (6 in our case). We have:

$$\mathbf{B} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 33.2226 & 0 & 0 \\ 0 & 16.5837 & 0 \\ 0 & 0 & 12.7714 \end{bmatrix}$$

$$\mathbf{AB} = \begin{bmatrix} 33.2226 & 0 & 0 \\ 0 & 16.5837 & 0 \\ 0 & 0 & 12.7714 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

This is sufficient to say that $\mathbf{Q}$ is full rank. Consequently, our system is controllable.

We can easily see that matrix $\mathbf{C}$ is a 6×6 identity matrix. While matrix $\mathbf{D}$ is a null 6×3. We can therefore calculate the matrix $\mathbf{N}$:

$$\mathbf{N} = \begin{bmatrix} \mathbf{C}^{\mathrm{T}} & \mathbf{A}^{\mathrm{T}}\mathbf{C}^{\mathrm{T}} & \cdots & \mathbf{A}^{\mathrm{T}^{n-1}}\mathbf{C}^{\mathrm{T}} \end{bmatrix} \tag{24}$$

Since $\mathbf{C}$ is a 6×6 identity matrix, $\mathbf{N}$ is full rank, which proves that our system is also observable.
We proved that the system is reachable and observable, so the Riccati equation is applicable.

8 Actuators

8.1 Reaction wheels

We define the matrix $\mathbf{A}$ as composed by 3 rows and as many columns as number of actuators. Each column represents the direction of the axis of rotation of the wheel. In our case, the matrix $\mathbf{A}$ corresponds to the identity matrix. As shown in [4], the model of the reaction wheel can be implemented using the following equations:

$$\mathbf{I}\underline{\dot{\omega}} + \underline{\omega} \wedge \mathbf{I}\underline{\omega} = \underline{T} + \underline{M_c} \tag{25}$$

$$\underline{\dot{h}_r} = \mathbf{I_r}\,\underline{\dot{\omega}_r} = \underline{M_r} \tag{26}$$

$$\underline{\dot{h}_r} = -\mathbf{A}^*(\underline{M_c} + \underline{\omega} \wedge \mathbf{A}h_r) \tag{27}$$

where $\mathbf{A}^*$ denotes the pseudoinverse of $\mathbf{A}$, $\underline{M_c}$ is the torque command computed by the attitude controller, $\underline{T}$ is the disturbance torque acting on the Cubesat, $\underline{\omega}$ is the spacecraft angular velocity, while $\underline{h_r}$ and $\underline{w_r}$ are respectively the vectors collecting the angular momentum and angular speed of each actuator about its spin axis (three actuators in our case). From Eq. (27), we can compute $\underline{\dot{h}_r}$ and, consequently, the wheel torque vector $\underline{M_r}$ required to achieve the commanded torque $\underline{M_c}$.

The CubeSat is equipped with three $NRWA-T005$ reaction wheels. From the datasheet [7], we can easily derive the following information:

- The reaction wheels inertia matrix is a 3×3 diagonal matrix, where every element of the diagonal is equal to $0.2 * 10^{-3}\,\mathrm{kg\,m^2}$.
- Each rotor is subjected to a maximum rotation rate of $250\,\mathrm{rad\,s^{-1}}$, which has been computed by dividing the maximum storable angular momentum for the moment of inertia of the reaction wheel.
- The maximum relative torque $\underline{M_r}$ applicable to each wheel is equal to $10^{-2}\,\mathrm{N\,m}$.
- Measurement of the wheels' speed is subjected to an error ranging between -1 to +1 revolutions per minute (that is, -0.1 to $+0.1$ radians per second). This uncertainty has been modelled using a Gaussian distribution with 0 mean value and a standard deviation equal to $\frac{0.1}{3}$. In this way, we can expect the error affecting each of the three components of w_r to remain within the $[-0.1, +0.1]$ interval with high probability (approximately 99.7% under a Gaussian assumption).

We must also take into account that the vector ω is subjected to error due to the non-idealities of the sensors. Consequently, the real control torque applied by the reaction wheels is given by:

$$\underline{M}_{c,real} = -\underline{\omega}_{real} \wedge \mathbf{A}\underline{h}_{r,real} - \mathbf{A}\dot{\underline{h}}_{r,real} \tag{28}$$

where $h_{r,real}$ is the vector collecting the real angular momentum of each actuator about its spin axis. That is, the angular momentum unaffected by the uncertainty of the wheels' speed measurement.

The reaction wheels model has been validated by confirming that the real torque $\underline{M}_{c,real}$ generated by the rotors is compatible with the torque command $\underline{M}_c$ computed by the attitude controller. As shown in Fig. 7, the reaction wheels manage to reproduce the command torque with an high degree of fidelity.

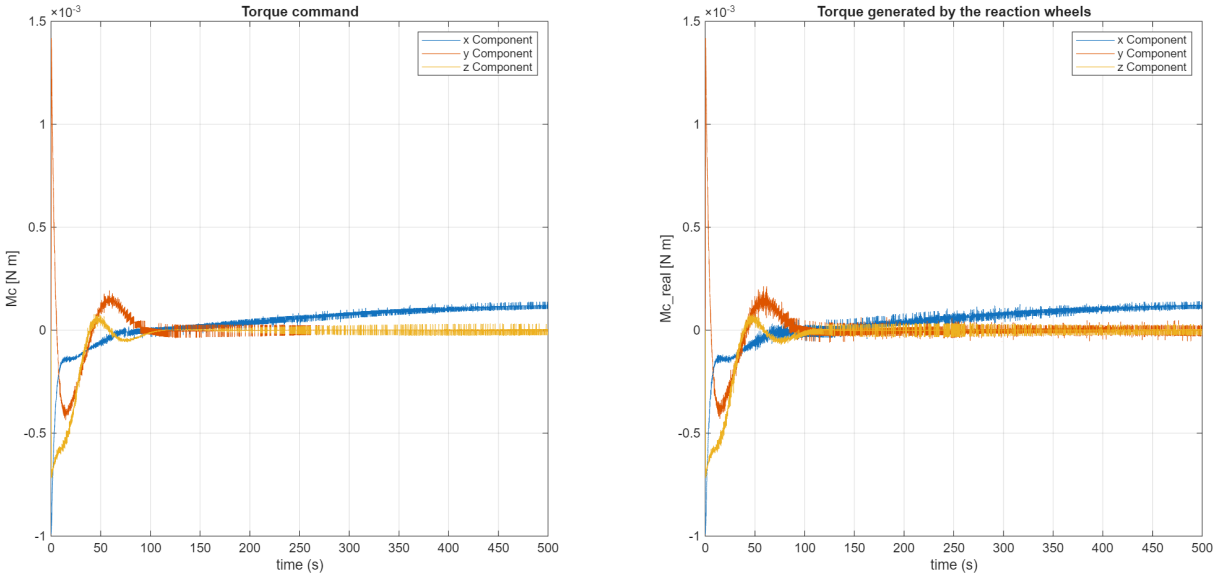


Figure 7: Comparison between torque computed by the attitude controller and torque generated by the rotors.

8.2 Magnetic torquer

The CubeSat is equipped with a three-axis magnetorquer which, during detumbling, takes as input the torque command computed by controller and, consequently, generates a local magnetic dipole in order to damp the angular velocity. The control torque $\underline{M}_c$ generated by the magnetic torquer cannot have any component parallel to the magnetic field $\vec{B}$. Consequently, the torque command is decomposed in a component parallel and orthogonal to the measured local magnetic field vector:

$$\underline{M}_{c,orth} = \underline{M}_c - \frac{(\underline{B}_m \cdot \underline{M}_c)\underline{B}_m}{\underline{B}_m^2} \tag{29}$$

The local magnetic dipole is generated so that it is orthogonal to the measured magnetic field:

$$\underline{m} = \frac{\hat{\underline{B}}_m \wedge \underline{M}_{c,orth}}{\underline{B}_m} \tag{30}$$

Consequently, the torque that is applied to the spacecraft is equal to:

$$\underline{M}_{c,real} = \underline{m} \wedge \underline{B}_r \tag{31}$$

where $\underline{B}_r$ represents the real local magnetic field.

As reported in [8], the magnetic torquer is unable to generate a local magnetic dipole moment exceeding $0.65\,\mathrm{A\,m^2}$ along the x and y axes. Along the z axis, the maximum achievable local magnetic dipole moment is $0.50\,\mathrm{A\,m^2}$.
Moreover, each component of the commanded magnetic dipole vector $\vec{m}$ is affected by a relative error reasonably assumed to lie within $\pm 10\%$ of the corresponding true value. This uncertainty is modelled by three independent normal distributions (one per component), each with zero mean and standard deviation equal to $\frac{1}{30}$ times the absolute value of the corresponding component. With this choice, the relative error on each component is expected to lie within $\pm 10\%$ with a probability of approximately 99.7% under a Gaussian assumption.

The magnetic torquer model has been validated in two ways.

- We have confirmed that the angle between the generated torque and the real magnetic field vector is constantly equal to 90 degrees (Fig. 8) .
- We have assumed the real magnetic field vector and the measured magnetic field vector to be constant in body frame and equal to [0;0;1]. As expected in this situation, the magnetic torquer has replicated with high fidelity the x and y components of the command torque computed by the attitude controller. While the third component of the generated torque is constantly null because of the fact that the magnetic torquer cannot generate a torque having any component parallel to the magnetic field vector (Fig. 9).

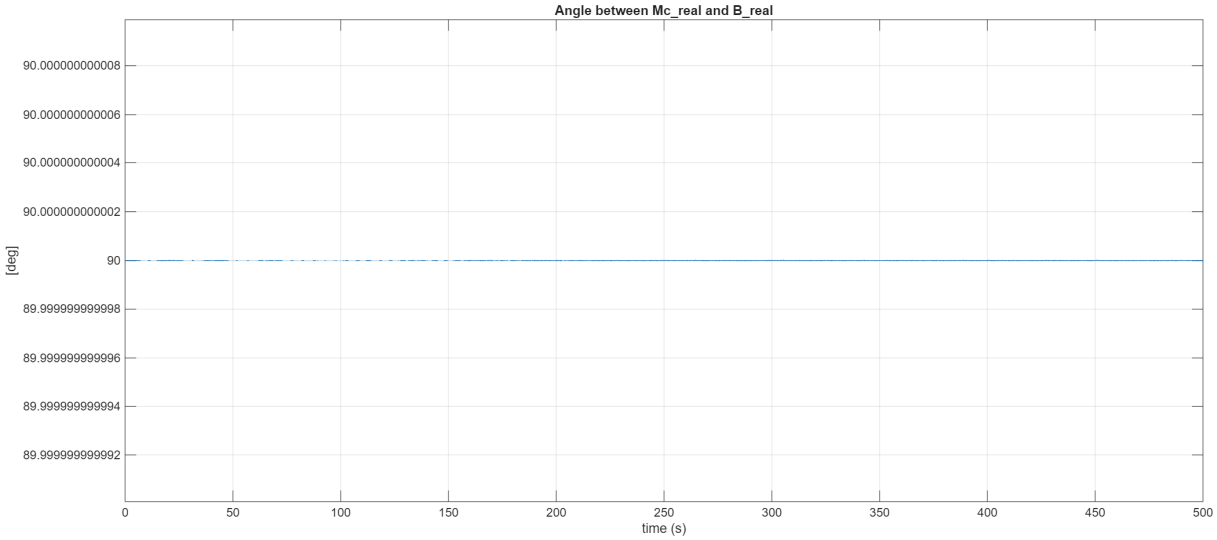


Figure 8: Angle between the real torque generated by the magnetic torquer and the real magnetic field vector.

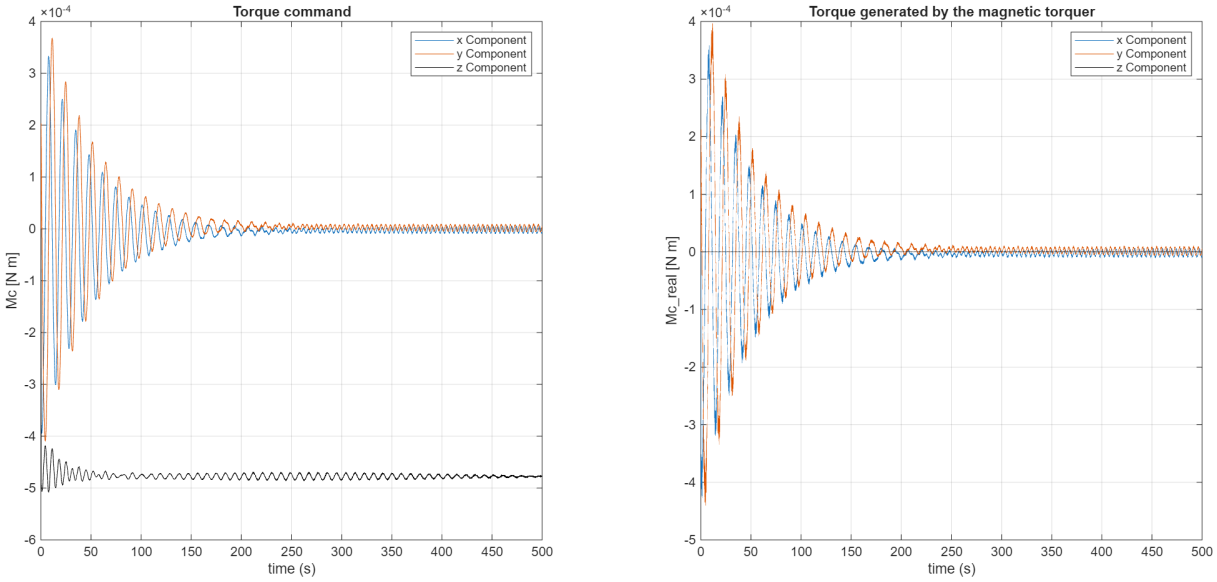


Figure 9: Comparison between torque computed by the attitude controller and torque generated by the magnetic torquer

9 Statistical analysis

We will now show that the system satisfies the mission pointing requirement in many different cases. The simulation is propagated for one orbit, which corresponds to a time equal to 5801 s.

- **Case 1**. Initial angular velocity vector in body frame: $[0.5; -0.3; 0.2]$ rad/s. Initial $\mathbf{A}_{B/N}$ matrix: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Initial true anomaly: $\theta_0 = 0°$

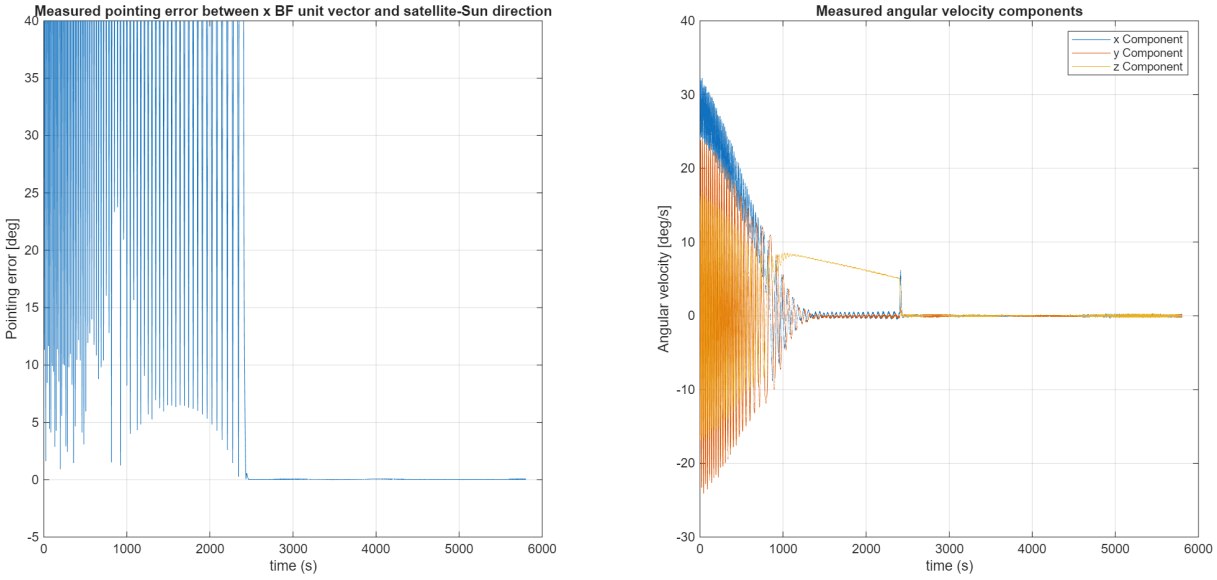


Figure 10: Plots of pointing error and angular velocity components expressed in body frame throughout the simulation in case 1

Tracking phase begins at around 2500 seconds. The mission objective of a pointing

error below 5 degrees is satisfied from the beginning of the tracking phase until the end of the simulation.

- **Case 2** Initial angular velocity vector in body frame: $[-0.2; 0.4; -0.1]$ rad/s. Initial $\mathbf{A}_{B/N}$ matrix: $\begin{bmatrix} 0.8666 & -0.5 & 0 \\ 0.5 & 0.8666 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Initial true anomaly: $\theta_0 = 30°$

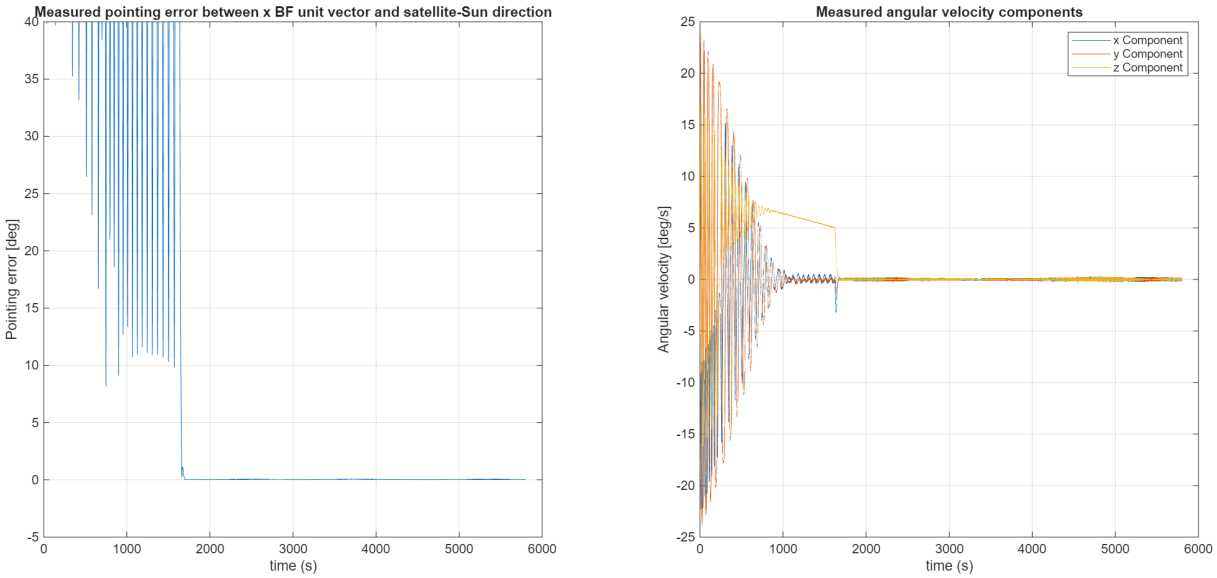


Figure 11: Plots of pointing error and angular velocity components expressed in body frame throughout the simulation in case 2

Tracking phase beginning at around 1800 seconds. Pointing requirement satisfied from the beginning of the tracking phase until the end of the simulation.

- **Case 3** Initial angular velocity vector in body frame: $[0; 0; 0.65]$ rad/s. Initial $\mathbf{A}_{B/N}$ matrix:
$\begin{bmatrix} 0.5 & 0 & 0.8666 \\ 0 & 1 & 0 \\ -0.8666 & 0 & 0.5 \end{bmatrix}$. Initial true anomaly: $\theta_0 = 90°$

Tracking phase beginning at around 2400 seconds. Pointing requirement satisfied from the beginning of the tracking phase until the end of the simulation.

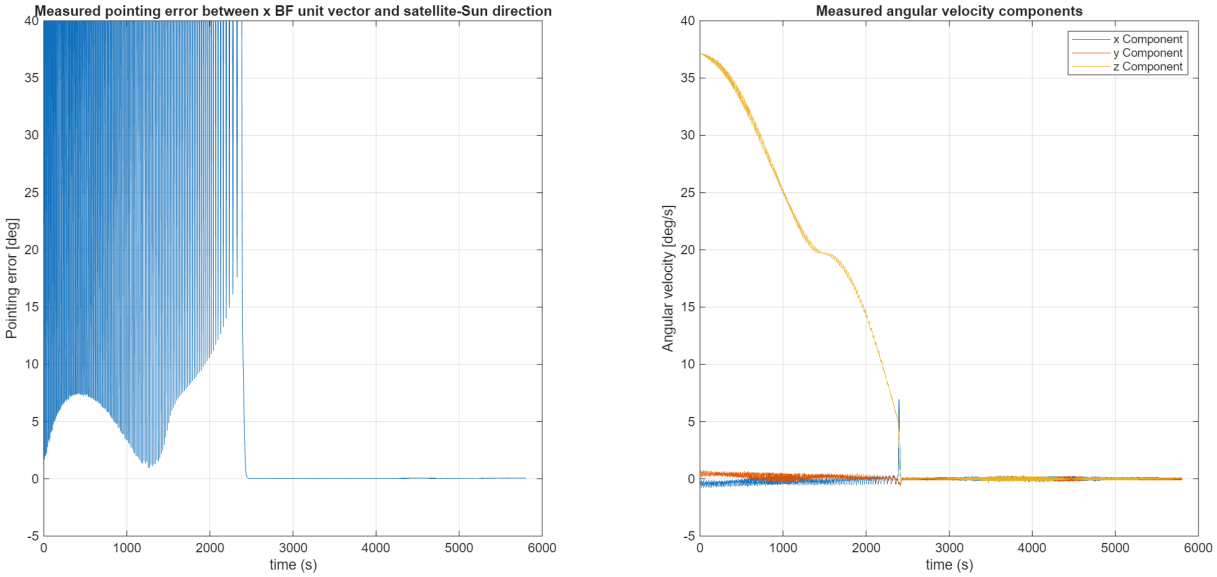


Figure 12: Plots of pointing error and angular velocity components expressed in body frame throughout the simulation in case 3

- **Case 4** Initial angular velocity vector in body frame: $[0.4; 0; -0.3]$ rad/s. Initial $\mathbf{A}_{B/N}$ matrix: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.7071 & -0.7071 \\ 0 & 0.7071 & 0.7071 \end{bmatrix}$. Initial true anomaly: $\theta_0 = 120°$

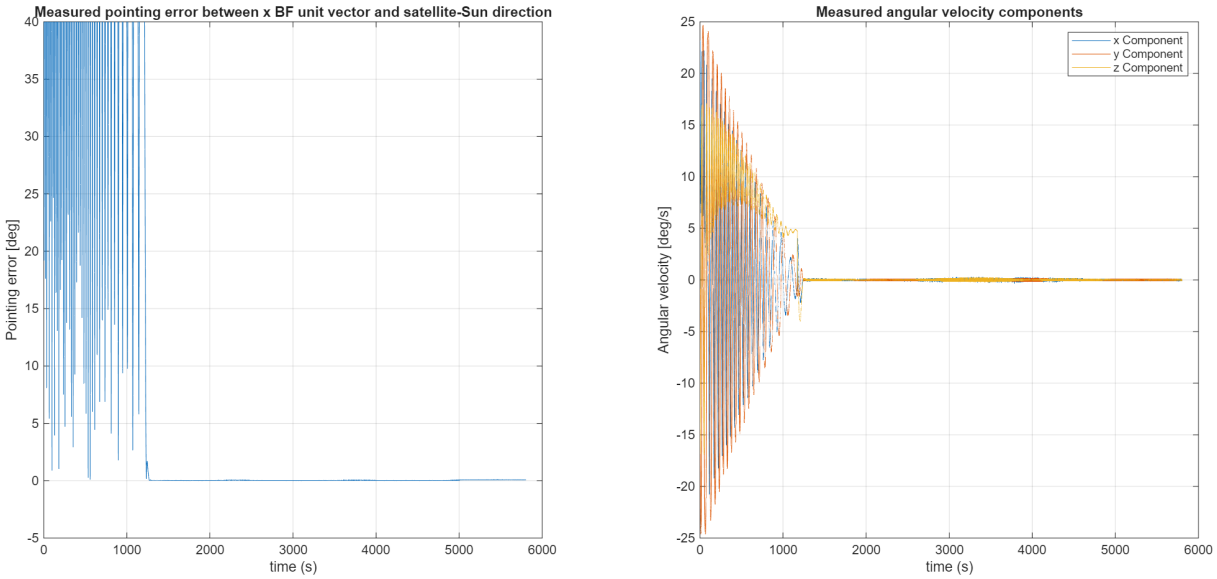


Figure 13: Plots of pointing error and angular velocity components expressed in body frame throughout the simulation in case 4

Tracking phase beginning at around 1200 seconds. Pointing requirement satisfied from the beginning of the tracking phase until the end of the simulation.

- **Case 5** Initial angular velocity vector in body frame: $[-0.3; 0.2; -0.3]$ rad/s. Initial $\mathbf{A}_{B/N}$ matrix: $\begin{bmatrix} -0.2475 & 0.4999 & 0.8300 \\ -0.5845 & -0.7602 & 0.2835 \\ 0.7727 & -0.4149 & 0.4803 \end{bmatrix}$. Initial true anomaly: $\theta_0 = 150°$

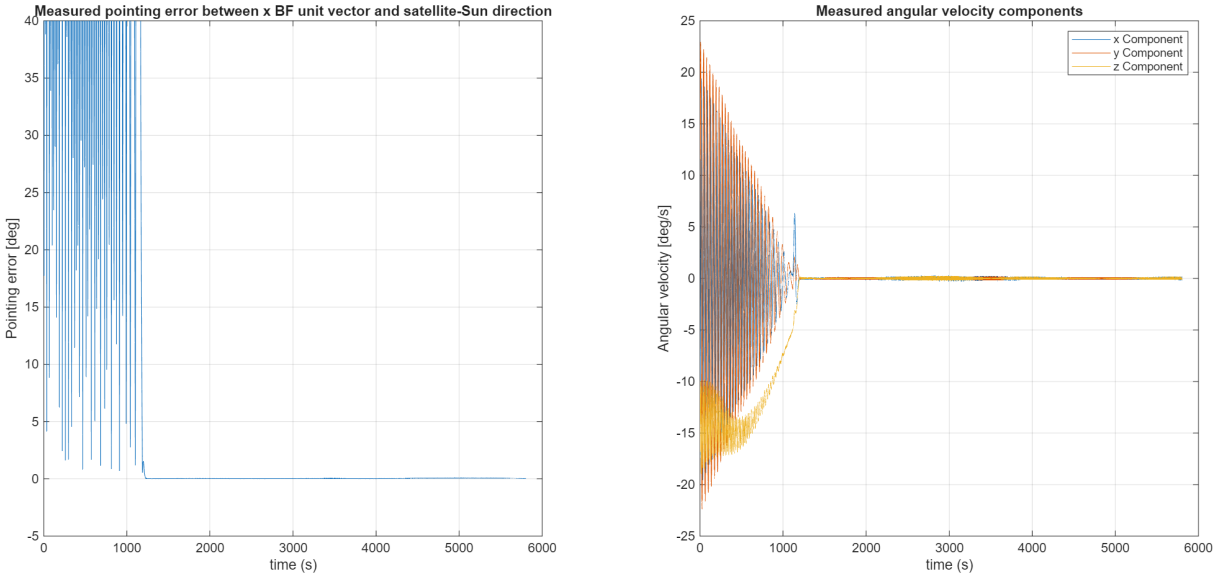


Figure 14: Plots of pointing error and angular velocity components expressed in body frame throughout the simulation in case 5

Tracking phase beginning at around 1200 seconds. Pointing requirement satisfied from the beginning of the tracking phase until the end of the simulation.

- **Case 6** Initial angular velocity vector in body frame: $[0.4; -0.5; 0]\,\text{rad/s}$. Initial $\mathbf{A}_{B/N}$ matrix: $\begin{bmatrix} -0.1431 & 0.0178 & -0.9896 \\ 0.7087 & -0.6960 & -0.1150 \\ -0.6908 & -0.7178 & 0.0870 \end{bmatrix}$. Initial true anomaly: $\theta_0 = 180°$

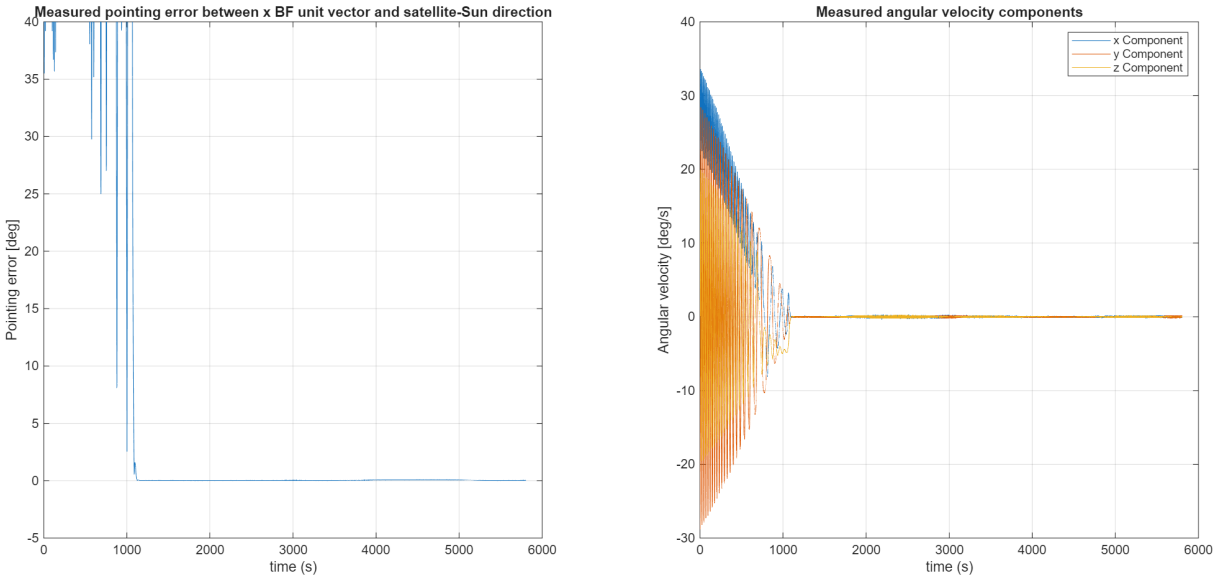


Figure 15: Plots of pointing error and angular velocity components expressed in body frame throughout the simulation in case 6

Tracking phase beginning at around 1100 seconds. Pointing requirement satisfied from the beginning of the tracking phase until the end of the simulation.

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